

Chapter 19: Properties of a Magnetic Field

Textbook Exercise Solutions & Answers



1. Write the appropriate terms in the blanks:

(a) The alloys called and are used for making industrial magnets.

alnico, Nipermag

(b) A magnetic field can pass through and

cardboard, bottle

(c) The intensity of a magnetic field is indicated by the lines of

force

(d) The real test of a magnet is

repulsion

2. With whom should I pair up?

Column 'A' Column 'B' 1. Compass a. The highest magnetic force 2. Door of a cupboard b. Like poles
3. Repulsion c. A magnet 4. Magnetic pole d. A magnetic needle

Column 'A'	Column 'B'
1. Compass	d. A magnetic needle
2. Door of a cupboard	c. A magnet
3. Repulsion	b. Like poles
4. Magnetic pole	a. The highest magnetic force

3. Write answers to the following questions:

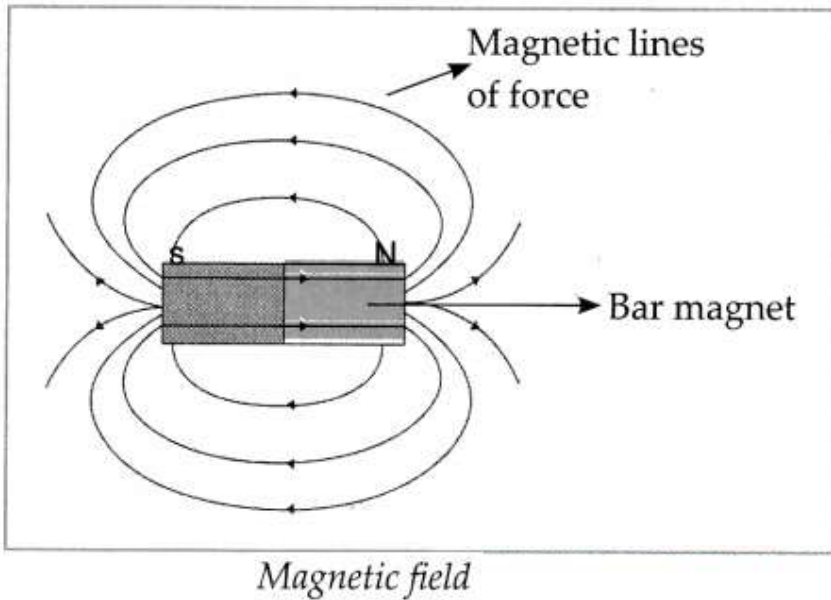
(a) Distinguish between the two methods of making artificial magnets

Single touch method	Double touch method
1. In this method, one magnet is used to magnetise the steel bar.	1. In this method, two magnets are used to magnetise the steel bar.
2. In this method, the magnet is dragged from one end of the steel bar to another end.	2. In this method, two magnets are dragged from the middle of the steel bar to the opposite directions.
3. As only one magnet is used to induce magnetism, this method is called single touch method.	3. As two bar magnets are used to induce magnetism, this method is called double-touch method.
4. The magnetism created by this method is of low strength and lasts for a short time.	4. The magnetism created by this method is of high strength and lasts for a longer time.

(b) Which substances are used for making electromagnets?

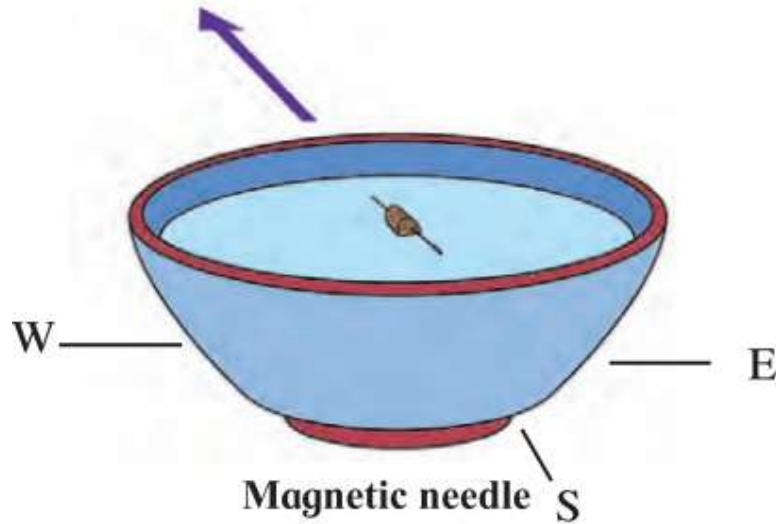
The substances used for making electromagnets are, an iron nail, insulated copper wire, one or more batteries and a pair of wire strippers.

(c) Write a note on magnetic field



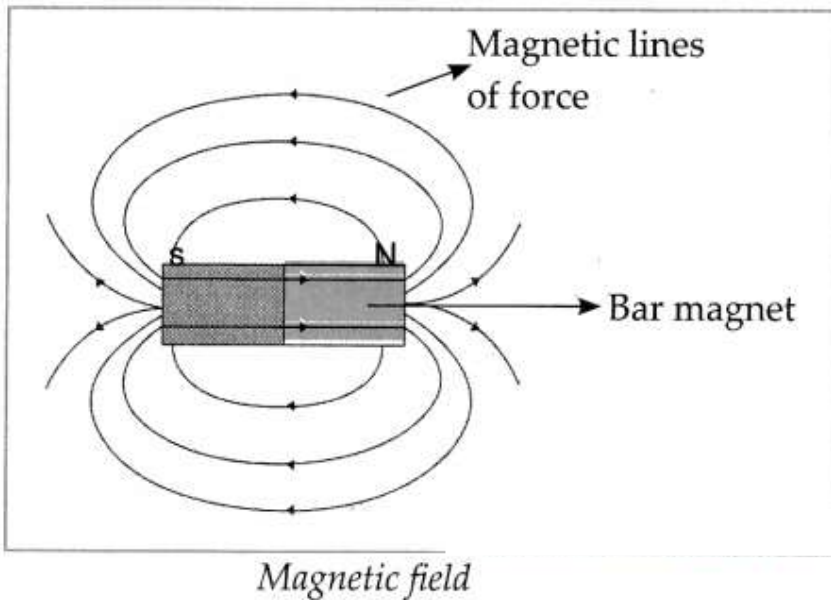
1. The region around a magnet where the magnetic force acts on an object is called a magnetic field.
2. The magnetic field around a magnet can be shown by means of magnetic lines of force.
3. The intensity of the magnetic field can be obtained from the number of lines of force.
4. The intensity of a magnetic field is low where the lines of force are sparse and the intensity is high where the lines of force are concentrated.
5. A magnetic field is the magnetic effect of electric currents and magnetic materials.
6. The magnetic field at any given point is specified by both the direction and the magnitude.
7. Magnetic fields are produced by electric current.

(d) Why is a magnetic needle used in a compass?



1. The magnetic compass is the most familiar compass type.
2. It functions as a pointer to magnetic 'north' the local magnetic meridian, because the magnetized needle at its heart aligns itself with the horizontal component of the earth's magnetic field.
3. A compass is an instrument used for navigation and orientation that shows direction relative to the geographic cardinal direction.

(e) Explain with the help of a diagram how the intensity and direction of the magnetic field of a bar magnet can be determined.



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4. Give detailed information about how the merchants of olden times used a magnet while

travelling.

1. The merchants of older times used a magnet whose north-south side was known.
2. When the magnet is freely suspended, the north pole of the magnet settles in the direction of the geographic north pole of the earth.
3. This is how the merchants come to know the directions while travelling.
4. The idea for using loadstone in a compass first appeared in China.
5. The first compass were made of loadstone, a naturally magnetised ore of iron.
6. The Chinese used “south pointing spoon” which when thrown upon the ground comes to rest pointing at the south.
7. Then the transition was made from loadstone spoons to iron needles.
8. In the medieval period a magnetic needle in the shape of a fish, floating in the bowl of water was used as a compass, pointing south.
9. It was called floating compass.
10. In the twentieth century, liquid filled magnetic compass began to appear.
11. There is a lesser known compass called dry mariners compass.